

Yanru Chen

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Research interests: Computer architecture; hardware/software co-design; heterogeneous memory; processing-in-memory and in-storage processing; AI inference and graph analytics.

EDUCATION

University of California San Diego Sep 2024 – Present
Ph.D. candidate in Electrical and Computer Engineering **Anticipated June 2027**
Advisor: Prof. Tajana Rosing; qualifying examination passed March 2026; GPA: 3.6/4.0.

Tsinghua University Aug 2021 – Jun 2024
M.S. in Electronic Information (Intelligent Manufacturing); GPA: 3.78/4.0.

Jilin University Aug 2017 – Jun 2021
B.E. in Electronic Science and Technology; College Honors; GPA: 3.53/4.0; rank: 6/108.

INDUSTRY EXPERIENCE

SK hynix Jun – Sep 2026
Emerging Memory AI Infrastructure Intern, AI Memory Solution / SOLAB San Jose, CA

GLAIM: agentic AI architecture and simulation *First author; submitted to ASPLOS 2027*

- Developed a trace-driven, event-driven simulator for agentic AI, integrating LLM prefill/decode, CPU tools, near-memory retrieval, persistent state and interconnect contention.
- Modeled hierarchical scheduling and compatible-request batching with dependency tracking, state ownership/versioning, copy-on-write sharing and data-readiness checks.
- Built a reproducible collection/replay workflow across five agentic and retrieval benchmarks; validated dependency ordering, resource reservations and data delivery in a portable simulator release.

RESEARCH EXPERIENCE

Systems Energy Efficiency Lab, UC San Diego Feb 2025 – Present
Graduate Researcher; advisor: Prof. Tajana Rosing La Jolla, CA

- Research in computer architecture, processing-in-memory/storage, heterogeneous memory, graph analytics, hyperdimensional computing and LLM inference.
- Shared-first-author work on CHIME and RAPID-Graph (DATE 2026) and HDDB (DAC 2026); second-author PALUTE (ISLPED 2026); first-author GEN-Graph manuscript submitted to IEEE TCAD.

Flexible and Printed Electronics Lab, UC San Diego Sep 2024 – Jan 2025
Memristor fabrication and multimodal sensing La Jolla, CA

Studied memristor fabrication and a multimodal sensing glove, including MEMS/NEMS fabrication and biomedical signal processing.

Micro-Electro-Mechanical System Laboratory, Tsinghua University Oct 2021 – Jun 2024
Flexible electronics and semiconductor sensors Beijing, China

- Wearable acetone gas sensor for breath detection and analysis; semiconductor gas sensing and microfabrication (Oct 2021 – Jun 2024).
- Laser-induced graphene flexible strain sensors and microfabrication (Sep 2022 – Nov 2023).

Computational Medicine Laboratory, Western University Jan – Aug 2020
CT perfusion analysis and machine learning London, Canada

Analyzed perfusion heterogeneity using PM3, fractal dimensions and machine learning; reconstructed 3D perfusion maps from CT images.

CONFERENCE PUBLICATIONS

* Equal contribution.

- [1] **Yanru Chen***, Runyang Tian*, Yue Pan, Zheyu Li, Weihong Xu, and Tajana Rosing. “CHIME: Chiplet-based Heterogeneous Near-Memory Acceleration for Edge Multimodal LLM Inference.” *Design, Automation and Test in Europe Conference (DATE)*, 2026.
- [2] Quanling Zhao*, **Yanru Chen***, Runyang Tian, Sumukh Pinge, Weihong Xu, Augusto Vega, Steven Holmes, Saransh Gupta, and Tajana Rosing. “HDDb: Efficient In-Storage SQL Database Search Using Hyperdimensional Computing on Ferroelectric NAND Flash.” *The Chips to Systems Conference (DAC)*, 2026.
- [3] **Yanru Chen***, Zheyu Li*, Keming Fan, Runyang Tian, John Hsu, Weihong Xu, Minxuan Zhou, and Tajana Rosing. “RAPID-Graph: Recursive All-Pairs Shortest Paths Using Processing-in-Memory for Dynamic Programming on Graphs.” *Design, Automation and Test in Europe Conference (DATE)*, 2026.
- [4] Runyang Tian, **Yanru Chen**, Weihong Xu, and Tajana Rosing. “PALUTE: Processing-In-Memory Acceleration via Lookup Table for Edge LLM Inference.” *IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*, 2026.
- [5] **Yanru Chen**, Jiaqi Liu, Yixin Liu, and Min Zhang. “Enhanced Strain Resistance of Fractal Fiber Laser-Induced Graphene for Flexible Electrodes Via Annealing and Plasma Etching.” *23rd International Conference on Solid-State Sensors, Actuators and Microsystems (Transducers)*, 2025, pp. 1109-1112. Poster presented by Yanru Chen.
- [6] Yixin Liu*, **Yanru Chen***, Zhibiao Wang, and Min Zhang. “Enhanced Electrical Conductivity in Laser-Induced Graphene-Silicon Carbide Laminated Nanosheets for Flexible Strain Sensors and Pulse Wave Velocity Assessment.” *IEEE 37th International Conference on Micro Electro Mechanical Systems (MEMS)*, Austin, TX, USA, 2024, pp. 919-922. Poster presented by Yanru Chen.
- [7] **Yanru Chen**, Sanjay R. Kharche, Yan Min Zhang, G. H. Janssen, and Christopher W. McIntyre. “Quantifying microvascular alterations due to a pharmacological agent.” *42nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Montreal, QC, Canada, 2020. Oral presentation by Yanru Chen.
- [8] Sanjay R. Kharche, **Yanru Chen**, and Christopher W. McIntyre. “Fractal Dimension Based Texture Analysis of CT Perfusion Imaging.” *42nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Montreal, QC, Canada, 2020. Oral presentation by Yanru Chen.

MANUSCRIPTS SUBMITTED TO ASPLOS 2027

- [1] **Yanru Chen**, Po-Kai Hsu, Sungwon Cho, Kiseok Lee, Yue Pan, Runyang Tian, Ashkan Moradifrouzabadi, Jungmin Choi, Mingu Kang, Shimeng Yu, Hoshik Kim, and Tajana Rosing. “GLAIM: Glass-Panel Full-Stack System for Agentic AI Inference with Heterogeneous M3D Memory.” *Submitted to ASPLOS 2027*.
- [2] Ashkan Moradifrouzabadi, Po-Kai Hsu, Seongwoo Yang, Kiseok Lee, **Yanru Chen**, Sungwon Cho, Hyunsoo Bang, Jinwoo Choi, Tajana Rosing, Jaesik Lee, Shimeng Yu, Jungmin Choi, Hoshik Kim, and Mingu Kang. “Co-Designing Compute-Interleaved 3D Memory and Adaptive Chunk Scheduling for MoE Serving.” *Submitted to ASPLOS 2027*.

JOURNAL PUBLICATIONS

- [1] **Yanru Chen**, Yixin Liu, Jiaqi Liu, Yuzhen Li, Yuhan Liu, Wenjie Zhang, Liuyang Han, Dongkai Wang, Shuhong Cao, Hanxiao Liu, and Min Zhang. “Porous PDMS-ZnO Wearable Gas Sensor for Acetone Biomarker Detection and Breath Analysis.” *ACS Applied Materials & Interfaces*, 2024.
- [2] Jiaqi Liu, Yuan Yu, Rujun Zhang, **Yanru Chen**, Yanlong Guo, Yi Zhang, Ran Tao, Jing-Ting Luo, Hairong Zheng, Pingfa Feng, and Min Zhang. “Trapping nanoscale particles via quasi-Scholte mode in acoustofluidics.” *Lab on a Chip*, 2025.
- [3] Yuhan Liu, Liuyang Han, Siqi Lv, Tao Jiang, Mingkai Duan, Hanyu Guo, Yuzhen Li, Qisen Xie, **Yanru Chen**, Dongkai Wang, and Min Zhang. “Emotional and Directional Enabled Programmable Flexible Haptic Interface for Enhanced Cognition in Disabled Community.” *Research*, 2025.
- [4] Qisen Xie, Liuyang Han, Jie Liu, Wenjie Zhang, Liuyan Zhao, Yuhan Liu, **Yanru Chen**, Yuzhen Li, Qian Zhou, Ying Dong, and Min Zhang. “Kirigami-Inspired Stretchable Piezoelectret Sensor for Analysis and Assessment of Parkinson’s Tremor.” *Advanced Healthcare Materials*, 2025.
- [5] Yuzhen Li, Yunfei Wang, Yixin Liu, Yuhan Liu, Liuyang Han, **Yanru Chen**, Juntian Qu, Puxiang Lai, and Xiang Qian. “Simultaneously encapsulation and formation of PDMS-MWCNTs composites for multi-directional microchannel force sensors.” *IEEE Sensors Journal*, 2024.

OTHER SUBMITTED MANUSCRIPTS AND PREPRINTS

- [1] **Yanru Chen**, Runyang Tian, Zheyu Li, Mahbod Afarin, Weihong Xu, and Tajana Rosing. “GEN-Graph: Heterogeneous PIM Accelerator for General Computational Patterns in Graph-based Dynamic Programming.” *Submitted to IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*.
- [2] **Yanru Chen**. “Low-Cost Spin-Coating Process for SnO₂-Based Memristors: Comparative Study of Different Solvent Systems and Device Performance.” *SSRN 5208743*, 2025. *Preprint*.

EARLIER INDUSTRY EXPERIENCE

Shenzhen Hanit Industrial Technologies Co. <i>Assistant Engineer, Department of Hardware</i> Developed flexible strain sensors using LiG–SiC nanosheets with self-temperature compensation.	Jun – Oct 2023 Shenzhen, China
Vivolight Medical Device & Technology Co. <i>Algorithm Intern, Department of Algorithm</i> Worked on OCT imaging integrating near-infrared spectroscopy with machine-learning-based lipid imaging.	Apr – May 2021 Shenzhen, China

AWARDS AND SCHOLARSHIPS

UCSD CSE Best Doctoral Research Award	Jun 2026
DAC Young Fellow Award	Mar 2026
Second Best Oral Presentation, Workshop on Learning and Information Theory	Nov 2024
Jacobs Fellowship, University of California San Diego	Aug 2024
Outstanding Undergraduate Graduation Project Award, Jilin University	Jun 2021
Excellent Student of the College (Top 5%), Jilin University	2018 – 2021
MITACS Globalink Research Internship Award, Western University	Feb 2020

TECHNICAL SKILLS

Programming: Python, C++, MATLAB, Verilog.
EDA and simulation tools: Altium Designer, Cadence Virtuoso, COMSOL, AutoCAD.
Platforms and microcontrollers: Arduino, ESP32, FPGA (Xilinx/Altera).